MILITARY SPECIFICATION

GRENADE XM74,
HEMISPHERES FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of the Male and Female Hemispheres for the XM74 Grenade.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6866 - Inspection, Penetrant Method of
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9298784 - Female Hemisphere
9298785 - Male Hemisphere
9313567 - Disc, Tungsten
9313626 - Tungsten, Powder Alloy

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E8 - Tension Testing of Metallic Materials
ASTM E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B311 - Density of Cemented Carbides

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings, specifications and standards.

3.1.1 Tungsten powder. - The powder shall be hydrogen reduced tungsten powder produced by normal production methods.

3.1.2 Alloying elements. - Elemental or prealloyed powders for alloying shall be high purity (99.5% by weight minimum).

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3.2. Parts. - All parts shall be in accordance with the applicable drawings, specifications and standards.

3.2.1 Disc

3.2.1.1 Microstructure. - The microstructure shall exhibit a uniform distribution of tungsten and binder material and no matrix depletion (porosity) when viewed at 200 magnifications and tested as specified in 4.5.3.

3.2.1.2 Physical quality. - Material shall be pressed as one piece with no sinter brazing or other joining methods permitted.

3.3 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. - All parts shall be fabricated in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined. Scoring shall be full, complete and undamaged over the required area.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of defects. - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

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4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Female Hemisphere	9298784	50
Male Hemisphere	9298785	50
Disc, Tungsten	9313567	30*

* See 6.4.a

4.3.2 Inspection to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63343(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Grenade XM74, Hemisphere and Disc for				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Female Hemisphere (Dwg. 9298784)				
	Examination for Defects	50	-	3.2	4.4.2.2
	Male Hemisphere (Dwg. 9298785)				
	Examination for Defects	50	-	3.2	4.4.2.3
	Disc, Tungsten (Dwg. 9313567)				
	Examination for Defects (excluding 102, 103, 104, 106)	20	-	3.2	4.4.2.1
	Examination for Defects (102, 103, 104, 106 only)	10(a)	-	3.2	4.4.2.1

NOTES:
(a) Samples to be taken from group of 20 discs.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formations. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each blend of powder mix shall constitute a lot.

4.4.1.1 Traceability to sintered lots. - The contractor shall establish a system satisfactory to the procuring agency of maintaining traceability to sintered lots.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63343(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Disc, Tungsten		9313567
			NEXT HIGHER ASSEMBLY
			9298784, -5

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Density incorrect		0.40%	3.2	4.5.2
102	Hardness incorrect	10	0-1	3.2	4.5.1
103	Ultimate tensile strength incorrect	10	0-1	3.2	4.5.4
104	Elongation incorrect	10	0-1	3.2	4.5.4
105	Scoring incomplete, interrupted, or missing		0.40%	3.2	Visual
106	Microstructure incorrect	10	0-1	3.2.1.1	4.5.3
Minor					
201	Overall diameter incorrect		0.65%	3.2	Gage
202	Diameter of scored area incorrect		0.65%	3.2	Gage
203	Frag size incorrect		0.65%	3.2	Gage
204	Thickness to groove incorrect		0.65%	3.2	Gage
205	Total thickness incorrect		0.65%	3.2	Gage
206	Bottom radius greater than .015		0.65%	3.2	Gage
207	Angle between frags incorrect		0.65%	3.2	Gage
208	Angle between frag and bottom surface		0.65%	3.2	Gage
209	Flatness incorrect (3 places)		0.65%	3.2	Gage
210	Evidence of poor workmanship (excluding 105)		1.00%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63343(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Hemisphere, Female		9298784
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Surface cracks or deep cuts		100%	3.2	4.5.5
102	Spherical radius incorrect		0.40%	3.2	Gage
103	Inner diameter incorrect		0.40%	3.2	Gage
Minor					
201	Outer diameter incorrect		0.65%	3.2	Gage
202	Depth of lip incorrect		0.65%	3.2	Gage
203	Chamfer incorrect		0.65%	3.2	Gage
204	Flat area is beyond permitted dimension		0.65%	3.2	Gage
205	Flatness incorrect		0.65%	3.2	Gage
206	Evidence of poor workmanship		1.00%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-63343(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Hemisphere, Male	1 OF 1	9298785
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Surface cracks or deep cuts		100%	3.2	4.5.5
102	Spherical radius incorrect		0.40%	3.2	Gage
103	Inner diameter incorrect		0.40%	3.2	Gage
Minor					
201	Outer diameter incorrect		0.65%	3.2	Gage
202	Depth of lip incorrect		0.65%	3.2	Gage
203	Chamfer incorrect		0.65%	3.2	Gage
204	Flat area is beyond permitted dimension		0.65%	3.2	Gage
205	Flatness incorrect		0.65%	3.2	Gage
206	Evidence of poor workmanship		1.00%	3.4	Visual

NOTES:

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4.4.3 Testing. - Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Hardness. - The test shall be performed in accordance with ASTM E18. If any test specimen fails to meet the requirement, it shall be classed defective and the lot shall be rejected. (Destructive test).

4.5.2 Density. - The test shall be performed in accordance with ASTM B311 using methods and equipment satisfactory to the procuring agency. If any disc fails to meet the requirement, it shall be classed defective. (Non-destructive test).

4.5.3 Microstructure. - The test shall be performed on a test blank from a disc using any method satisfactory to the procuring agency. If any test specimen fails to meet the requirement, it shall be classed defective and the lot shall be rejected. (Destructive test).

4.5.4 Mechanical properties. - Tensile test blanks conforming to ASTM E8, Figure 6, shall be prepared from completed discs. These samples shall be free of scoring when subjected to testing at a strain rate of .02 per minute as specified in ASTM E8. If any test specimen fails to meet ultimate tensile strength and elongation requirements, it shall be classed defective and the lot shall be rejected. (Destructive test).

4.5.5 Penetrant inspection. - Testing shall be performed as specified in MIL-I-6866, Table I. Test equipment and procedures shall be satisfactory to the procuring agency. If any hemisphere fails to meet the requirement, it shall be classed defective and removed from the lot. (Non-destructive test).

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5. PACKAGING

5.1 Preservation and packaging. - Preservation and packaging shall be in accordance with best commercial practices.

5.2 Marking. - Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The components covered by this specification are intended for use in the XM74 Grenade.

6.2 Ordering data. - See MIL-S-48078.

6.3 Submission of inspection equipment for design approval. - See MIL-A-48078. Submit designs as required to: Commander, US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Informational tests. - The following shall be submitted for informational testing only to: Commander, US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND, ATTN: DRDAR-SCM-P:

a. From First Article samples, 10 Discs, Dwg. #9313567

b. 2 Discs per production lot, Dwg. #9313567.

c. One Female or Male Hemisphere per production lot, Dwg. #9298784 or #9298785.

d. 200 grams Tungsten Powder Alloy per powder manufacturer's lot, Dwg. #9313626 (Blended powder with no wax)

e. 100 grams each alloy ingredient per powder manufacturer's lot, Dwg. #9313626 (Elemental or prealloyed powder)

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6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading "US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)" may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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